to an eleven-dimensional universe containing two-dimensional membranes.

For technical reasons, Witten first came upon the eleventh dimension in his studies of the strong coupling properties of the Type I IA string, and there the story is quite similar. As in the Heterotic-E example, there is an eleventh dimension whose size is controlled by the Type IIA coupling constant. When its value is increased, the new dimension grows. As it does, Witten argued, the Type IIA string, rather than stretching into a ribbon as in the Heterotic-E case, expands into an "inner tube," as illustrated in Figure 12.8. Once again, Witten argued that although theorists have always viewed Type IIA strings as one-dimensional objects, having only length but no thickness, this view is a reflection of the perturbative approximation scheme in which the string coupling constant is assumed to be small. If nature does require a small value of this coupling constant then it is a trustworthy approximation. Nevertheless, Witten's arguments and those of other physicists during the second superstring revolution do give strong evidence that the Type IIA and Heterotic-E "strings" are, fundamentally, two-dimensional membranes living in an eleven-dimensional universe.

But what is this eleven-dimensional theory? At low energies (low compared to the Planck energy), Witten and others argued, it is approximated by the long-neglected eleven-dimensional supergravity quantum field theory. But for higher energies, how can we describe this theory? This topic is currently under intense scrutiny. We know from Figures 12.7 and 12.8 that the eleven-dimensional theory contains two-dimensional extended objects—two-dimensional membranes. And as we shall soon discuss, extended objects of other dimensions play an important role as well. But beyond a hodgepodge of properties, no one knows what this eleven-dimensional theory is. Are membranes its fundamental ingredients? What are its defining properties? How does it purport to make contact with physics as we know it? If the respective coupling constants are small, our best current answers to these questions are described in previous chapters, since at small coupling constants we are led back to the theory of strings. But if the coupling constants are not small, no one currently knows the answers.